



Fly Away: Lifting Fault Security through Canaries and the Uniform Random Fault Model

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Abstract. Cryptographic implementations are vulnerable to active physical attacks where adversaries inject faults to extract sensitive information. Existing fault models, such as the threshold and random fault models, assume limitations on the amount or probability of injecting faults. Such models, however, insufficiently address the case of practical fault injection methods capable of faulting a large proportion of the wires in a circuit with high probability. Prior works have shown that this insufficiency can lead to concrete key recovery attacks against implementations proven secure in these models. We address this blind spot by introducing the *uniform random fault model*, which relaxes assumptions on the amount/probability of faults and instead assumes a uniform probabilistic faulting of all wires in a circuit or region. We then show that security in this new model can be reduced to security in the random fault model by inserting *canaries* in the circuit to ensure secret-independent fault detection. We prove that combining canaries with a more classical fault countermeasure such as redundancy can lead to exponential fault security in the uniform random fault model at a polynomial cost in circuit size in the security parameter. Finally, we discuss the interactions between our work and the practical engineering challenges of fault security, shedding light on how the combination of state-of-the-art countermeasures may protect against injections of many high probability faults, while opening a path to methodologies that formally analyze the guarantees provided by such countermeasures.

Keywords: Fault attack · Uniform random fault model · Canary circuits

1 Introduction

Modern electronic communication frequently relies on cryptographic algorithms to obtain security guarantees like confidentiality, integrity or authenticity of transmitted information. Symmetric or asymmetric cryptosystems are responsible for exchanging secret data like messages or keys, establishing secure communication over insecure channels. Yet, whenever a concrete physical manifestation of a cryptographic algorithm, say a hardware implementation or software executed on a hardware device, needs to operate on secret material in presence of untrusted parties — or even under direct physical exposure to malicious adversaries — implementation attacks become a critical concern.

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1.1 State of the Art and Related Work

Among physical attacks, we distinguish passive and active attacks, also known as side-channel [KJJ99] and fault injection [BS97, BDL97] analysis, respectively. The former is typically based on closely observing an implementation’s behavior or emissions during execution of an algorithm, while the latter physically interacts or tampers with the computing hardware, introducing faulty behavior to reveal secret data. Both analysis types have been demonstrated to lead to devastating attacks on unprotected implementations in practice [KJJ99, JT12]. Naturally, in order to provide cost-effective protection against such powerful attackers, it is crucial to precisely understand their capabilities. Commonly, the goal in theoretical implementation security research is to develop an abstract adversary model which reflects the capabilities of the considered class of attackers in such a way that (i) it allows to formally argue at reasonable complexity about the security in presence of adversaries and (ii) it ensures that implementations proven secure in the model are resistant to practical attacks.

1.1.1 Passive Attacks

The cryptographic research community has addressed this challenge with considerable success regarding passive adversaries over the past decades. Abstractions such as the probing model [ISW03], the noisy leakage model [PR13], the random probing model [DDF14], the bounded moment model [BDF⁺17] and the robust probing model [FGP⁺18] have been proposed and studied over time, together with their security reductions to establish important relationships between each other. Compositional strategies have even been introduced in order to prove security properties for small sub-function implementations individually (so-called gadgets) which may then be composed arbitrarily into larger circuits which automatically inherit the designated security properties [BBD⁺16, CS20]. Despite questions with respect to the validity of the independence and noise assumptions made by these models that will probably never be completely eliminated, there exist established techniques to minimize the reliance on these assumptions [FGP⁺18, CGLS21, MMMS23, BCM⁺23]. Overall, theoretical progresses towards analyzing side-channel security helped to delineate the border between (increasingly weak) physical assumptions, that must be fulfilled by cryptographic engineers, and the security guarantees that can be formally analyzed. In other words, they helped restrict and clarify engineering efforts so that leakage-resilient designs can become more efficient and easier to analyze, thanks to a good separation of duties.

1.1.2 Active Attacks

In the area of active physical adversaries, the situation is arguably less clear. Abstract models to capture the capabilities of adversaries include the threshold fault model [IPSW06], the consolidated fault model [RSG23], the random fault model [FPV11, DK12, DN23] (which we will call the κ -random fault model in the following) and the general random fault model [BFG⁺24]. However, each of these models has known limitations originating from the need to make assumptions which may not always hold in practice (see Figure 1). Security reductions or other formal relationships have not been established yet either.

The threshold fault model (and its generalization as the consolidated fault model) allows the injection of a limited number of high-precision faults. Security in this model can be obtained for example through redundancy-based countermeasures which are well-suited to detect or correct a bounded number of faults injected into the computation [SM12]. Once this threshold is successfully overcome by an adversary, all security promises are void. This naturally begs the question whether overcoming a predetermined bound on the maximum number of units the adversary may fault is actually hard. Recent research indicates that this is not necessarily the case. For example, it has been demonstrated

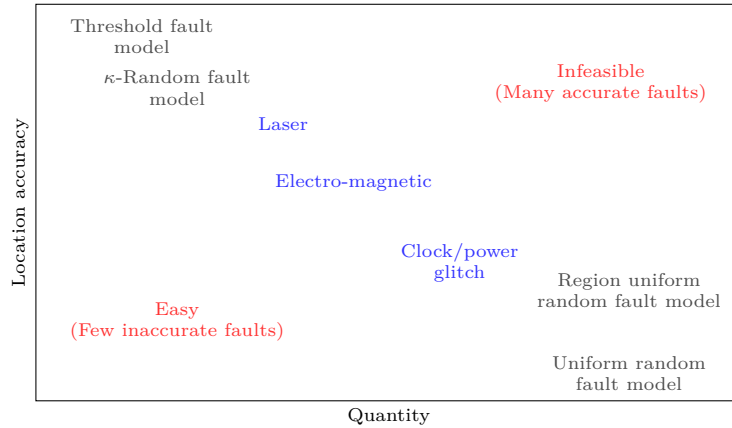


Figure 1: Classification of fault models and common fault injection techniques with respect to quantity and location accuracy of the considered faults.

that redundancy-based countermeasures proven secure in the threshold fault model for small threshold values (which lead to efficient implementations) are comparatively easy to break in practice [BBM⁺22]. In theory, increasing the threshold is a viable option to strengthen the resistance. However, the security vs. performance trade-off might still be unsatisfactory as higher threshold choices incur harsh cost or performance penalties while there is no theoretical (or practical) upper limit on the number of faults an adversary may produce simultaneously.

The κ -random fault model (of which the general random fault model is a generalization) is a probabilistic fault adversary model and conceptually allows the injection of an arbitrary number of faults. In this model, the success of each injection is upper bounded by a certain probability instead. This makes the number of injected faults scale with the amount of computation,¹ avoiding one of the major limitations of the threshold fault model. The κ -random fault model has however its own downsides, a major one being the difficulty of finding a reasonable fault probability bound κ . Indeed, there is no clear upper bound on κ since some attack methods may fault many wires in a circuit at once with high probability [BCN⁺06].² On the other hand, if κ is set to a high value, designing systems with strong provable security becomes infeasible or prohibitively expensive, similar to choosing a too high upper bound in threshold-based models.

Beyond the abstractions discussed above, some unbounded fault models have also been proposed in the literature. Examples include [IPSW06] which, among others, considers unbounded stuck-at-0 faults which is quite specific, or [GLM⁺04] which allows arbitrary faults in a tamperable memory, therefore excluding the faults on computation that we consider. This last model is also considered in the first paper on non-malleable codes [DPW18], which gave rise to numerous extensions with various restrictions on the type of functions that can be applied – see [FMVW14, FHMV17, BFO⁺20, BFMV22] for a few examples.

1.2 Our Contributions

In this work, we introduce a new fault adversary model called the *uniform random fault model* which neither puts a threshold on the amount nor on the probability of injections.

¹Here, the amount of computation refers to both dimensions, space (e.g., size of a circuit) and time (e.g., number of cycles), equally.

²As a trivial theoretical example, switching off the global power supply of the device would fault a large fraction of wires. Clock and power glitching attacks provide a similar capability in practice, with some degree of control on the amount of faulted wires.

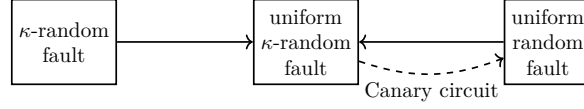


Figure 2: Technical relationships between selected fault models. An arrow $x \rightarrow y$ indicates that security in model x implies security in model y .

The intended goal is to model fault injection methods that produce a large quantity of faults with relatively high probability, as such attacks have been shown to bypass state-of-the-art provably secure countermeasures in practice [BBM⁺22]. Instead of limiting the quantity of injected faults, our model restricts the *location accuracy* of injections, i.e., limiting the ability to decide which wires get faulted, by assuming a uniform probabilistic faulting of all wires in a circuit. While it is obviously feasible to inject a large number of high probability faults in practice (e.g., through a clock or power glitch), it is commonly considered hard to do so with high location accuracy. Figure 1 depicts a classification of the discussed fault models and a few selected injection techniques with respect to quantity and location accuracy of faults. As illustrated, the threshold and random fault models represent the injection of small quantities of faults with high location accuracy, while our new model falls into the opposite corner representing large quantities of produced faults with low accuracy. Therefore, it fills a concrete gap and extends the provable security coverage of possible adversarial strategies. We also show that the uniformity assumption (and therefore the accuracy restriction) can be slightly relaxed with the *region uniform random fault model* where the adversary has some (limited) control over the fault locations. Adversary models combining a large quantity and high location accuracy would be most conservative, but hardly useful in practice as imposing less and less restrictions on adversaries makes it harder and harder to prove security of implementations beyond minimal toy circuits.

As a second important contribution we establish the reductions between fault models shown in Figure 2: alongside the κ -random fault model and the uniform random fault model, the uniform κ -random fault model is the natural intersection of these two models, leading to trivial relationships. More importantly, we introduce the notion of *canary circuits*, a circuit countermeasure based on fault detection which essentially consists in inserting “tripwire” circuitry that is completely unrelated to the sensitive data, and reduces uniform random fault security to uniform κ -random fault security. This is achieved by imposing a trade-off on the adversary: either inject few faults and stay within the guarantees of the base (uniform) κ -random fault security circuit, or inject many faults and get detected by the canary. The unique characteristic of canary circuits, compared to other fault detection schemes, is that they are not connected to the secret computation. This comes with the big advantage that canary circuits are immune to (statistical) ineffective fault attacks [Cla07, DEK⁺18, DEG⁺18], and therefore adding canary circuits never degrades threshold or κ -random fault security. The flip side of this immunity is that canary circuits only improve security in the uniform random fault model: a threshold or κ -random fault adversary may simply choose to not fault the canary circuits.

Finally, we introduce a (theoretical) circuit compiler³ that turns unprotected circuits into ones that are secure in all three fault models. We achieve threshold fault security using duplication, add duplicated error correction (majority vote) circuits to limit fault propagation and thus tolerate a fixed number of faults per protected gate which ensures κ -random fault security. Ultimately, the addition of canary circuits provides uniform random fault security.

³For completeness we mention that our formal analysis considers only stateless arithmetic circuits. Yet, those are equivalent to stateful circuits considering execution within a fixed or bounded number of clock cycles, which is often the case for practical cryptographic circuits.

1.3 Outline

We introduce our new modular security models in Section 2, including security games that describe the goal of the adversary and the environment. We then explain the canary countermeasure in Section 3, prove that it reduces uniform random fault security to κ -random fault security. Next, we propose an abort mechanism implementation that prevents computation outputs to be exposed to the adversary when canary circuits are triggered in Section 4. Then, in Section 5, we consider a relaxation of the uniformity assumption with the region uniform random fault model. Finally, we conclude the work in Section 6.

2 Preliminaries and Fault model

2.1 Notations

We use calligraphic letters (e.g. $\mathcal{A}$) to denote sets and algorithms, capital letters (e.g. A) for random variables and circuits, and sans-serif capitals (e.g. $\mathbf{A}$) to denote probability distributions. For any discrete set $\mathcal{A}$, $\mathbb{D}\mathcal{A}$ denotes the set of probability distributions over $\mathcal{A}$. We denote by id the identity function $\mathbb{F}_2 \rightarrow \mathbb{F}_2$. Given two random variables X and Y over a discrete domain $\mathcal{D}$, the statistical distance is $\Delta(X, Y) = \frac{1}{2} \sum_{d \in \mathcal{D}} |\Pr[X = d] - \Pr[Y = d]|$.

2.2 Circuit model

We consider Boolean circuits modeled as directed acyclic graphs $C = (\mathcal{G}, \mathcal{W})$ whose vertices $\mathcal{G}$ are gates and whose edges $\mathcal{W}$ are wires (we denote $n_w = |\mathcal{W}|$). There are a few kinds of gates: input gates, output gates, randomness gates, logic gates and copy gates. All gates have fan-out 1, except for the output gates (fan-out 0) and copy gates (arbitrary fan-out), while the input and randomness gates have fan-in 0, copy gates and output gates have fan-in 1, and logic gates have arbitrary fan-in: typically 0 (for constants), 1 or 2. The evaluation e_C of a circuit C is a probabilistic function $\mathbb{F}_2^{n_i} \rightarrow \mathbb{F}_2^{n_o}$ mapping the values of the input gates x to the values of the output gates $e_C(x)$.

2.3 Fault injection

We model fault injections as changing the value of wires, by means of a fault function $f : \mathbb{F}_2 \mapsto \mathbb{F}_2$ which transforms the *correct* value of a wire into a *faulted* value. This model therefore includes the bit-flip, stuck-at, random value and “no fault” cases. A faulting $F \in \mathcal{F}$ of a circuit C is a collection $(f_w)_{w \in \mathcal{W}}$ containing a fault function for each wire of the circuit. A faulted evaluation $e_C^F(x)$ proceeds similarly to a normal evaluation, but applies the fault function before assigning the value to a wire. Finally, we will consider probabilistic faultings $\mathbf{F}$, which are formally probability distributions over all possible faultings $\mathcal{F}$ of a circuit: $\mathbf{F} \in \mathbb{D}\mathcal{F}$. A faulted evaluation with a probabilistic faulting $e_C^{\mathbf{F}}(x)$ is a two-step process: a faulting F is first sampled from $\mathbf{F}$, then the result is $e_C^F(x)$.

Remark Some other works model faults as changing gates and not wires (see e.g. [FGS23]), which is a more general model than ours when copy gates are included. We decided to work with a wire fault model thanks to its simplicity: a fault injection is modeled by a single probabilistic $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ function. In contrast, a gate fault model would be more complex, defining one probabilistic function for each gate type, which makes it hard, if not impossible, to apply our notion of uniformity in the following. Also, this complexity often leads to limiting the circuit model in seemingly arbitrary ways (e.g., limiting possible gates to a subset of all possible $\mathbb{F}_2^2 \rightarrow \mathbb{F}_2$ gates in [FGS23]). Our model includes copy gates which allows considering fault injections that affect gate input and faults affecting a gate output.

2.4 Random fault models

The definition of probabilistic faulting is very broad and very powerful: an adversary that can inject an arbitrary faulting in a circuit is able to reprogram the circuit to perform almost⁴ any function of its choice. We therefore introduce *fault models* as ways to restrict the control of the adversary over the injected faults.

A first restriction is independence: we assume that the probabilistic faulting distribution F is the product of independent probabilistic faultings for each wire of the circuit. Further, in the κ -random fault model introduced in [DN23], each wire can only be faulted with probability at most κ :

Definition 1 (κ -random fault model). A faulting F is a κ -random faulting if there exists $(F_1, \dots, F_{n_w})$ such that $F = F_1 \times \dots \times F_{n_w}$ and for all $i = 1, \dots, n_w$, $\Pr_{F_i \leftarrow F_i}[F_i \neq \text{id}] \leq \kappa$.

Let us now introduce the new uniform fault model, where instead of restricting the probability of successfully injecting a fault, the adversary cannot precisely target individual wires. We achieve this by forcing all individual wire faultings to be identical.

Definition 2 (Uniform random fault model). A faulting F is a uniform random faulting if there exists $(F_1, \dots, F_{n_w})$ such that $F = F_1 \times \dots \times F_{n_w}$ and $F_1 = \dots = F_{n_w}$.

In this model, the only freedom is the selection of a single-wire probabilistic faulting, which amounts to specifying an (asymmetric) binary communication channel, characterized by the probability p_{01} of the fault $0 \rightarrow 1$ and the probability p_{10} of the fault $1 \rightarrow 0$, which we next denote as a (p_{01}, p_{10}) -exact uniform random faulting. Practically, the adversary can control the kind of faults it injects, and the probability of injecting them, but does not control on which wires the faults get injected. We relax this restriction in Section 5 with the introduction of fault regions.

Finally, we may combine both restrictions: a uniform κ -random faulting is a probabilistic faulting which is both a κ -random faulting and a uniform random faulting. Concretely, this imposes $p_{01} \leq \kappa$ and $p_{10} \leq \kappa$.

2.5 Fault security games

We consider a circuit to be secure against a probabilistic faulting F if the output of the faulted evaluation can be simulated from the output of a non-faulted evaluation. This implies that faulting the circuit does not reveal new information about the secret circuit input x .

Definition 3 (Fault security). A circuit C is ϵ -fault secure under a probabilistic faulting F if there exists a probabilistic simulator algorithm $\mathcal{S}$ such that for any $x \in \mathbb{F}_2^{n_i}$, $\Delta(e_C^F(x), \mathcal{S}(e_C(x))) \leq \epsilon$.

We also use another security definition that includes the notion of *abort signals*. Indeed, in many cases, a circuit may detect the presence of an injected fault, and we may wish to then stop all computations.

Definition 4 (Fault security with abort). Let C be a circuit whose outputs $y \in \mathbb{F}_2^{n_o}$ are partitioned into $(a, y') \in \mathbb{F}_2^{n_a} \times \mathbb{F}_2^{n_o - n_a}$ for some $n_a \in \mathbb{N}$ and there exists $t^a \in \mathbb{F}_2^{n_a}$ such that for all non-faulted evaluations, $a = t^a$. C is ϵ -fault secure with abort under a probabilistic faulting F if there exists a probabilistic simulator algorithm $\mathcal{S}$ such that for any $x \in \mathbb{F}_2^{n_i}$

$$\Delta(\mathcal{O}(e_C^F(x)), \mathcal{S}(e_C(x))) \leq \epsilon$$

where $\mathcal{O}((a, y')) = y'$ iff $a = t^a$ and $\mathcal{O}((a, y')) = \perp$ otherwise.

⁴It can effectively change the function of each logic gate by faulting its inputs and/or outputs. Looking at 2-input gates, any XOR gate can be exchanged for an XNOR gate and vice-versa, while gates that are not XOR nor XNOR can be transformed into any other.

This abort capability, besides being useful for security proofs, can also be directly implemented in practice by means of a reset circuitry. For such an implementation, a reset would be triggered by any of the abort signals (which can be a mix of active-high and active-low signals). Let us however note that this abort capability is not fundamentally changing the security model since, as we show in [Section 4](#), security without aborts can be reduced to security with aborts by adding an output-gating circuit ensuring that when an abort is triggered, all the outputs of the circuit are set to a public (e.g. all-0) value (an output-gating construction is given in [Section 4](#)).

Finally, let us remark that the definitions are concerned about the secrecy of (part of) the circuit inputs, which covers typical fault attacks (e.g. DFA, SIFA) against cryptographic implementations. A different requirement often analyzed [[DN23](#), [THN⁺24](#)] is circuit correctness. Such a security notion is harder to properly define since in our fault models, it is possible to fault an output wire with non-negligible probability, breaking “naive” correctness definitions.⁵ On the other hand, correctness without secret input (e.g. signature verification) can be easier to achieve than input secrecy, since ineffective attacks are not a concern.

3 Canary circuits

In this section, we introduce the *canary circuit* countermeasure whose goal is to detect the presence of fault injection attacks. The general idea is to add a part of the circuit that performs computations independently of the actual computation of interest. If this computation turns out to be incorrect, an abort signal will be activated. This forces the adversary to only inject few faults (i.e., high probability having the identity as the fault function), otherwise the canary circuit will be faulted, leading to an abort with high probability. Formally, we reduce security in the uniform random fault model to security in the κ -random fault model.

We consider a trivial example of canary circuit, which simply connects ℓ output gates to constant “0” gates and ℓ output gates to constant “1” gates as the abort signals. This canary uses a mix of active-low and active-high abort signals in order to ensure fault detection even in the presence of stuck-at-0 and stuck-at-1 faults.⁶

Definition 5 (Trivial canary compiler). Given a circuit C that outputs y , the ℓ -trivial canary compiler outputs a circuit identical to C , with the addition of ℓ constant “0” gates, each connected to a single wire, itself connected to a new output, and likewise for ℓ constant “1” gates. These new outputs are the abort signals a , such that the overall outputs of the circuit are (a, y) . The expected abort signal pattern t^a is $(0, \dots, 0, 1, \dots, 1)$.

Let us now prove that the trivial canary compiler reduces uniform random fault security with abort to κ random fault security. First, we analyze the effectiveness of the canary for fault detection as a function of the faulting.

Lemma 1. *Let C' be the output of the ℓ -trivial canary compiler for some circuit C , $\mathcal{O}$ be the abort fault security oracle and $\mathbf{F}$ be a (p_{01}, p_{10}) -exact uniform random faulting, then for any x , $\Pr[\mathcal{O}(e_{C'}^{\mathbf{F}}(x)) = \perp] = 1 - (1 - p_{01})^\ell (1 - p_{10})^\ell$.*

Proof. The probability that no active-high (resp. active-low) wire gets faulted is $(1 - p_{01})^\ell$ (resp. $(1 - p_{10})^\ell$), since all faults are independent, and each abort signal depends on only a single wire. Therefore, the probability of faulting at least one abort signal is $1 - (1 - p_{01})^\ell (1 - p_{10})^\ell$, which is equal to the probability of $\mathcal{O}$ outputting $\perp$. $\square$

⁵A more involved definition would have to change a bit the circuit model to allow verification of the output values, such as done in [[THN⁺24](#), Definition 9].

⁶This construction is arguably fairly theoretical. We discuss the practical instantiation of canary circuits in [Appendix A](#).

We can then show how the security with canary depends on the initial security of the compiled circuit and of the canary itself.

Theorem 1. *Let C' be the output of the ℓ -trivial canary compiler for some circuit C . If C is ϵ_κ -fault secure under any uniform κ -random faulting for any $\kappa \in [0, 1]$, then C' is ϵ' -fault secure with abort under any uniform probabilistic faulting with $\epsilon' = \max_{\kappa \in [0, 1]} (1 - \kappa)^\ell \epsilon_\kappa$.*

Proof. Let us consider a (p_{01}, p_{10}) -exact uniform random faulting. Let $\mathcal{S}$ be the simulator for the security of C . We build $\mathcal{S}'((a, y))$ as follows: output $\perp$ with probability $\alpha = 1 - (1 - p_{01})^\ell (1 - p_{10})^\ell$, and otherwise output $\mathcal{S}(y)$.

For any $x \in \mathbb{F}_2^{n_i}$ and $y \in \mathbb{F}_2^{n_o}$, we then have $\Pr[\mathcal{S}'(e_{C'}(x)) = y] = (1 - \alpha) \Pr[\mathcal{S}(e_C(x)) = y]$, and $\Pr[\mathcal{S}'(e_{C'}(x)) = \perp] = \alpha$. On the other hand, $\Pr[\mathcal{O}(e_{C'}^F(x)) = y] = (1 - \alpha) \Pr[e_C^F(x) = y]$, $\Pr[\mathcal{O}(e_{C'}^F(x)) = \perp] = \alpha$, thanks to [Lemma 1](#) and the independence of the canary circuit with respect to the rest of the circuit.

These observations lead to the following computation:

$$\begin{aligned} & \Delta(\mathcal{O}(e_{C'}^F(x)), \mathcal{S}'(e_{C'}(x))) \\ &= \frac{1}{2} \sum_{y \in \mathbb{F}_2^{n_o} \cup \{\perp\}} |\Pr[\mathcal{O}(e_{C'}^F(x)) = y] - \Pr[\mathcal{S}'(e_{C'}(x)) = y]| \\ &= \frac{1}{2} |\alpha - \alpha| + \frac{1}{2} \sum_{y \in \mathbb{F}_2^{n_o}} (1 - \alpha) |\Pr[\mathcal{O}(e_C^F(x)) = y] - \Pr[\mathcal{S}(e_C(x)) = y]| \\ &= (1 - \alpha) \Delta(\mathcal{O}(e_C^F(x)), \mathcal{S}(e_C(x))) \\ &\leq (1 - p_{01})^\ell (1 - p_{10})^\ell \epsilon_{\max(p_{01}, p_{10})} \end{aligned}$$

where the inequality comes from the observation that F is a $\max(p_{01}, p_{10})$ -random faulting. By universally quantifying over p_{01} and p_{10} , we get that C' is uniform fault ϵ' -secure with

$$\epsilon' = \max_{p_{01}, p_{10} \in [0, 1]} (1 - p_{01})^\ell (1 - p_{10})^\ell \epsilon_{\max(p_{01}, p_{10})} = \max_{\kappa \in [0, 1]} (1 - \kappa)^\ell \epsilon_\kappa$$

where, for the last equality, we observe that taking $\kappa = \max(p_{01}, p_{10})$ proves that the right-hand side is an upper-bound of the left-hand side, while taking $(p_{01}, p_{10}) = (\kappa, 0)$ proves the reversed inequality. $\square$

Corollary 1. *Let C' be the output of the ℓ -trivial canary compiler for some circuit C . If C is ϵ -fault secure under any uniform κ -random faulting, then C' is $\max(\epsilon, (1 - \kappa)^\ell)$ -fault secure with abort under any uniform probabilistic faulting.*

Proof. For any $\kappa' \in [0, 1]$, let $\epsilon_{\kappa'} = \epsilon$ if $\kappa' \leq \kappa$ and $\epsilon_{\kappa'} = 1$ otherwise, such that C is $\epsilon_{\kappa'}$ -fault secure under any uniform κ' -random faulting. It then suffices to prove that for any $\kappa' \in [0, 1]$, $(1 - \kappa')^\ell \epsilon_{\kappa'} \leq \max(\epsilon, (1 - \kappa)^\ell)$. If $\kappa' \leq \kappa$, this holds because $\epsilon_{\kappa'} = \epsilon$ and $0 \leq (1 - \kappa')^\ell \leq 1$. Otherwise, we have $\epsilon_{\kappa'} = 1$ and $(1 - \kappa')^\ell \leq (1 - \kappa)^\ell$. $\square$

3.1 Applying canary to redundant circuits

Let us now consider a toy fault countermeasure to demonstrate the application of canary circuits and [Theorem 1](#). Let C be an unprotected circuit made of gates with at most 2 inputs, C^r be the compilation of this circuit with replication and majority voting, and finally C^c be the ℓ -canary compilation of C^r . The design is further parameterized by n which is the number of replicas of each gate. We will show that C^c achieves exponential security with abort against any uniform random faulting at a polynomial increase of circuit size in the parameter n (for some well-chosen ℓ as a function of n).

In more detail, C^r is built as shown in [Figure 3](#). Each gate of C is replicated n times (we assume that n is an odd integer), and each of the n gate replicas has an error correction

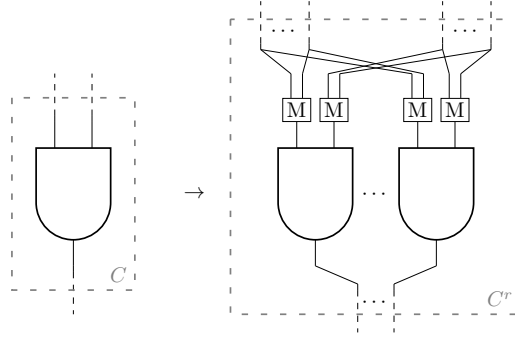


Figure 3: Compilation with replication of a circuit made of a single AND gate, “M” boxes denote error correction circuits that implement majority voting. Compilation of larger circuits simply connects together compilations of single gates.

circuit connected to each of its inputs. These error correction circuits take as input n wires and perform a majority vote to compute their single output (which is connected to the gate replica): if at least $\lceil n/2 \rceil$ wires are 1, then the output is 1, otherwise it is 0. Finally, if the output of a gate g is connected to an input of a gate g' in C , the n output wires of the gate replicas corresponding to g are connected to the n error correction circuits connected to the corresponding inputs of the replicas of g' in C^r . This construction ensures $\lceil n/2 \rceil$ -threshold fault security. The function computed by the error correction circuit is equivalent to summing all n input bits and comparing the sum to $\lceil n/2 \rceil$, which can be done in complexity linear in n using 2-input gates (e.g. by a tree of full adders). Let us assume that an error correction circuit contains αn wires for some $\alpha > 0$, then the number of wires in C^r is $|C^r| \leq |C| (2\alpha n + 1)n$ where $|C|$ is the number of gates in C (the factor 2 is the bound on the number of input wires per gate in C).

Let us now analyze the uniform κ -random faulting security of C^r . Looking at a replicated gate with its input correction circuits, let us assume that each replicated input is faulted on at most $\lceil n/2 \rceil$ wires. Then, for each of the n gate replicas, if there is no fault in the gate input/output wire, and no fault in the associated input error correction circuits, then the output of that replica is not faulted. Since there are $2\alpha n + 1$ wires in total in this part of the circuit ($2\alpha n$ for the two error correction circuits, one wire for the output of the gate replica), the output is faulted with probability at most $\kappa_r \leq (2\alpha n + 1)\kappa$ in the κ -random fault model. Therefore, there are strictly more than $\lceil n/2 \rceil$ faulted replicas with probability at most

$$\kappa_g = \sum_{k=\lceil n/2 \rceil}^n \binom{n}{k} \kappa_r^k (1 - \kappa_r)^{n-k} \leq \sum_{k=0}^{\lceil n/2 \rceil} \binom{n}{k} \kappa_r^{n-k} \leq 2^{n-1} \kappa_r^{\lceil n/2 \rceil}.$$

Iterating the condition of faulting at most $\lceil n/2 \rceil$ replicas for all gates in C , we get that C^r performs a correct evaluation⁷ with probability at least $|C| \kappa_g$. Therefore, C^r is $\epsilon_{r,\kappa}$ -secure in the κ -random fault model with

$$\epsilon_{r,\kappa} \leq |C| 2^{n-1} ((2\alpha n + 1)\kappa)^{\lceil n/2 \rceil}.$$

Next, we consider the canary compilation using [Theorem 1](#), giving that C^c is ϵ' -secure

⁷This ignores the issue of faults in the “output” majority vote circuits of C^r . We sketch a toy example solution: connect each duplicated output to a chain of error correction circuits duplicated n times on the outputs. If the chains are long enough, it becomes overwhelmingly likely that at least one of the circuits at each chain is not faulted, and therefore its output wires are all equal to the correct (public) output. All subsequent wires in the chain, until the final circuit output, can therefore be simulated while knowing only the correct output values.

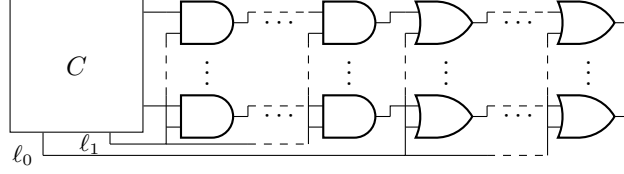


Figure 4: Abort compilation of a circuit C with ℓ_0 active-high and ℓ_1 active-low abort signals.

with abort in the uniform random fault model with

$$\begin{aligned} \epsilon' &= \max_{\kappa \in [0,1]} |C| 2^{n-1} ((2\alpha n + 1)\kappa)^{\lceil n/2 \rceil} (1 - \kappa)^\ell \\ &= |C| 2^{n-1} \left(\frac{2\alpha n + 1}{\ell / \lceil n/2 \rceil + 1} \right)^{\lceil n/2 \rceil} \left(\frac{1}{\lceil n/2 \rceil / \ell + 1} \right)^\ell \end{aligned}$$

(the maximum is reached for $\kappa = 1/(\ell / \lceil n/2 \rceil + 1)$). We can loosen this bound to simplify it to

$$\epsilon' \leq \frac{|C|}{4} \left(\frac{2(n+1)(2\alpha n + 1)}{\ell} \right)^{\lceil n/2 \rceil},$$

which shows that by taking ℓ a few times larger than $\approx 4\alpha n^2$, we get exponential security increase with n . The overall circuit size is then $2\ell + |C|n(2\alpha n + 1)$ wires (and the number of gates is within a factor 2 of the number of wires), hence the overhead of the canary circuit will be fairly small, given that $|C|$ is in practice much larger than the security parameters.

4 Abort-free circuits

In this section, we demonstrate that the uniform random fault model with abort is not fundamentally weaker than the uniform random fault model. We do this by exhibiting a compiler that, given a circuit secure in the uniform random fault model with aborts, generates a circuit with a similar security level in the uniform random fault model without abort. The core idea of this compiler is to “gate” the outputs of the circuit: if any abort signal is triggered, then all outputs are set to a constant value, ensuring that no information is leaked. This is achieved thanks to the circuit illustrated in Figure 4.⁸

Definition 6 (Abort compiler). Let C be a circuit with output (a, y) where $y \in \mathbb{F}_2^{n_o}$ is the main output and $a \in \mathbb{F}_2^{n_a}$ contains abort signals with non-faulted value t^a , and whose output a is not connected to the circuit input. The output of the abort compiler for C is a circuit C' identical to C , but with the output gates removed, and an *output gating* circuit is added, constructed as follows. First, each a_j wire is connected to a copy gate whose outputs are $a_{i,j}$ for $i = 1, \dots, n_o$. Then, let us rename each wire y_i as $z_{i,0}$. For each $i = 1, \dots, n_o$, we then build the following circuit: for each $j = 1, \dots, n_a$, we add a gate $z_{i,j} = g(z_{i,j-1}, a_{i,j})$ where g is a OR gate if $t_i^a = 0$ and g is a AND gate if $t_i^a = 1$. Finally an output gate z_i is connected to each z_{i,n_a} wire.

Theorem 2. Let C' the ℓ -trivial canary compilation of a C and let C'' be the abort compilation of C' . If C is uniform κ -random fault ϵ -secure, then C'' is uniform 0.5-random fault $\max(\epsilon, (1 - \kappa)^{2\ell})$ -secure.

⁸This construction demonstrates feasibility but is not optimized. We discuss more concrete instantiations of abort mechanism in Appendix A.

Proof. Let us consider a (p_{01}, p_{10}) -exact uniform random faulting F . If $p_{01} \leq \kappa$ and $p_{10} \leq \kappa$, then C'' is ϵ -secure. Indeed, we can perfectly simulate the computation of a and of the gates introduced by the abort compiler, leaving the simulation error on the outputs z bounded by the simulation error on y , which is ϵ using the uniform κ -random fault security of C .

Otherwise, let us assume without loss of generality that $p_{01} \geq \kappa$. Considering a single output of the circuit z_i , we distinguish the *gated* case $a_{i,*} \neq t^a$, which implies that the value of z_i depends only on $a_{i,*}$, from the *non-gated* case $a_{i,*} = t^a$, where $z_i = y_i$. The non-gated case happens if for each j , neither a_j nor $a_{i,j}$ are faulted, or a_j and $a_{i,j}$ are both faulted, giving

$$\Pr[a_{i,*} = t^a] = ((1 - p_{01})^2 + p_{01}p_{10})^\ell ((1 - p_{10})^2 + p_{10}p_{01})^\ell \leq (1 - \kappa)^{2\ell}$$

where the inequality uses the assumption that $\kappa \leq p_{01} \leq 0.5$ and $p_{10} \leq 0.5$. Considering all outputs z , the jointly non-gated case (i.e., any z_i is non-gated), denoted $\bar{G}$, happens with probability $\Pr[\bar{G}] \leq n_o(1 - \kappa)^{2\ell}$.

Next, we define the simulator $\mathcal{S}''$. It simulates the canary and gating circuit, and produces the corresponding output in the jointly gated case (i.e., all z_i 's are gated), while it outputs 0 otherwise. We denote by G' the jointly gated event for the simulator, which has the same probability as G . Further, for any x , conditioned on G and G' , the outputs of the faulted circuit $e_{C''}^F(x)$ and the simulator outputs $\mathcal{S}''(e_{C''}(x))$ are identically distributed. Finally, we compute

$$\begin{aligned} \Delta(e_{C''}^F(x), \mathcal{S}''(e_{C''}(x))) &= \frac{1}{2} \sum_{z \in \mathbb{F}_2^{n_o}} |\Pr[e_{C''}^F(x) = z] - \Pr[\mathcal{S}''(e_{C''}(x)) = z]| \\ &\leq \frac{1}{2} \sum_{z \in \mathbb{F}_2^{n_o}} |\Pr[e_{C''}^F(x) = z \wedge G] - \Pr[\mathcal{S}''(e_{C''}(x)) = z \wedge G']| \\ &\quad + \frac{1}{2} \sum_{z \in \mathbb{F}_2^{n_o}} |\Pr[e_{C''}^F(x) = z \wedge \bar{G}] - \Pr[\mathcal{S}''(e_{C''}(x)) = z \wedge \bar{G}']| \end{aligned}$$

where the first term is equal to 0, while the second term is a statistical distance multiplied by $\Pr[\bar{G}]$, which gives $\Delta(e_{C''}^F(x), \mathcal{S}''(e_{C''}(x))) \leq \Pr[\bar{G}] \leq n_o(1 - \kappa)^{2\ell}$, concluding the second case of the proof. $\square$

5 Region fault model

The uniform random fault model introduces the hypothesis that faults are injected uniformly over the whole circuit. This assumption may be too strong, e.g. if the adversary uses a laser with a small spot size, or if the circuit is executed over multiple clock cycles and the adversary can choose the timing of the fault injection. In this section, we therefore relax the uniformity requirement by introducing the region uniform random fault model, where the circuit is split into subsets named regions, and the adversary can inject a different random fault function in each region. In other words, the uniformity requirement now applies to subsets of the circuits instead of applying to the circuit as a whole.

Practically, defining regions amounts to characterizing the “location accuracy” of the adversary with a particular fault injection capability: the adversary can arbitrarily select the regions in which faults are injected, but not the wires within the regions. For example, for laser fault injection, a region will correspond to the wires under the laser spot, while for clock glitching attacks, the regions will be related to other characteristics such as the path delay.

Formally, a region R is a subset of the wires in a circuit C , and the regions of a circuit $\mathcal{R}$ (w.r.t. some adversarial capabilities) is a set of regions.

Definition 7 (Region uniform random fault model). Given a circuit C with regions $\mathcal{R}$, a region faulting selection is a tuple of disjoint⁹ regions $(R_1, \dots, R_n)$ where $R_i \in \mathcal{R}$ and a tuple of corresponding fault functions $(F_1^R, \dots, F_n^R)$. The corresponding region index of each wire $w \in \mathcal{W}$ is $i_w = \max\{i \in \{0, \dots, n\} \mid w \in R_i\}$, where $R_0 = \mathcal{W}$ (i.e., R_0 is the “default” region for wires in no other region).

A faulting F is a $\mathcal{R}$ -region uniform random faulting if there exists a region faulting selection such that $F = F_1 \times \dots \times F_{|\mathcal{W}|}$ where for every wire $w \in \mathcal{W}$, $F_w = F_{i_w}^R$, taking $F_0^R = \text{id}$.

The region uniform random fault model is therefore a relaxed version of the uniform random fault model, depending on the set of regions $\mathcal{R}$: it comes down to the uniform random fault model when there is only a single region that covers the whole circuit, while if there is a region for each individual wire, the model is equivalent to the very strong 1-random fault model. The canary countermeasure therefore still works in the region uniform random fault model if the canary circuit is inserted in such a way that each region contains enough canary tripwires, which notably requires that the regions are not too small.

Theorem 3. Let C' be the output of the trivial canary compiler for some circuit C with regions $\mathcal{R}$, such that ℓ active-high and ℓ active-low canary wires are inserted in each region $R \in \mathcal{R}$. If C is ϵ_κ -fault secure under any κ -random faulting for any $\kappa \in [0, 1]$, then C' is ϵ' -fault secure with abort under any $\mathcal{R}$ -region uniform probabilistic faulting with $\epsilon' = \max_{\kappa \in [0, 1]} (1 - \kappa)^\ell \epsilon_\kappa$.

Proof. The proof is similar to the proof of Theorem 1, but we have to consider a (p_{01}^i, p_{10}^i) -exact uniform random faulting for each of the regions $(R_i)_{i=1, \dots, n}$. We can then build a simulator that perfectly simulates the $\perp$ cases by outputting $\perp$ with probability $\alpha = 1 - \prod_{i=1}^n (1 - p_{01}^i)^\ell (1 - p_{10}^i)^\ell$, and we get (by the same computation as in the proof of Theorem 1)

$$\begin{aligned} \Delta(\mathcal{O}(e_{C'}^F(x)), \mathcal{S}'(e_{C'}(x))) &= (1 - \alpha) \Delta(\mathcal{O}(e_C^F(x)), \mathcal{S}(e_C(x))) \\ &\leq \prod_{i=1}^n (1 - p_{01}^i)^\ell (1 - p_{10}^i)^\ell \epsilon_{\max(\max_i p_{01}^i, \max_i p_{10}^i)} \end{aligned}$$

where $\mathcal{S}$ is the simulator for the circuit of C (remark that this is κ -random fault security, and not uniform κ -random fault security: the fault functions may differ across regions). By universally quantifying over p_{01}^i and p_{10}^i , we get that C' is $\mathcal{R}$ -region uniform random fault ϵ' -secure with $\epsilon' = \max_{\kappa \in [0, 1]} (1 - \kappa)^\ell \epsilon_\kappa$. $\square$

The construction of Section 3.1 also works in the region uniform random fault model thanks to Theorem 3, with the strengthened requirement to have enough canary wires in each region (i.e., a few times $4\alpha n^2$ canary wires per region, where n is the redundancy parameter). This is achievable (and has a reasonable cost) as long as the regions themselves contain at least a few times $4\alpha n^2$ wires, i.e., if they contain the equivalent of a few replicated gates.¹⁰

6 Conclusion

In this work we introduced a new formal fault adversary model called the *uniform random fault model* together with its variant called the *region uniform random fault model*. Both

⁹We assume disjoint regions for the sake of simplicity, since otherwise we have to specify the result of faulting multiple times a wire.

¹⁰In practice, it is therefore important that regions are well-characterized and are large enough. This may require the use of low-level countermeasures to limit the location accuracy of fault injection such as wire meshes [BCN⁺06]. We discuss these engineering challenges in more detail in Appendix A.

are designed to analyze the resistance to a previously ignored class of attackers that may inject a large quantity of high probability faults, though with limited location accuracy. We show that security in these new models can be reduced to security in the κ -random fault model by inserting canaries, a sort of “tripwire” circuitry that crucially does not compute over secret data. Based on these novel reductions we propose a circuit compiler that can be used to secure implementations in the threshold, random and (region) uniform random fault models jointly. Importantly, the resulting implementations provide exponential fault security in the (region) uniform random fault model at a polynomial cost in circuit size. Our models and constructions, for the first time, enable designers to formally argue about the security of implementations in presence of adversaries that may inject a large number of faults with high probability and therefore close a gap in current fault analysis literature.

Future work directions include the extension of the compiler to provide (region) uniform random fault security for concrete hardware circuits at a lower cost. Further, the practical application of our model should be evaluated by identifying and experimentally validating the relevant regions for various fault injection techniques. This would open the door to practical a instantiation of the compiler with optimized implementations of canary circuit and abort mechanism.

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A Engineering challenges

The requirements for canary circuits in the region uniform random fault model ([Theorem 3](#)) can be summarized as “complete-coverage and secret-independent fault-detection circuits, completed by a fault-resilient abort mechanism” ([Section 4](#)). These requirements obviously present a number of engineering challenges. However, we argue in the following that none of these challenges presents an insurmountable obstacle and that the practical research field has studied some of these problems for a long time already.

First, *complete-coverage*, i.e., covering all regions, practically means that canary circuits should be sensitive to all possible fault injection techniques. Defining cohesive fault regions therefore requires some insight about the way different fault injection methods affect certain subsets of logic elements or signals in a circuit simultaneously. This is an essential topic in fault characterization and has been an active field of research for decades, culminating in a recent batch of survey or Systematization of Knowledge (SoK) type research articles [[BH22](#), [BBB⁺23](#), [RSG23](#), [TNN24](#)]. As already mentioned in [Section 5](#), one of the classical examples giving rise to the idea of regions is laser fault injection attacks. These attacks not only allow precise timing of faults, but also precise spatial control of the injection. In this case, a selected region may consist of a certain number of logical units that are placed and routed in very close proximity to each other on a chip. The size of the region then depends on the laser spot size which is reaching the surface of the semiconductor device under attack, as well as the feature size of the manufactured integrated circuit technology. While both, large and small laser spots have been shown to lead to effective attacks in practice [[BBM⁺22](#)], they commonly represent different adversarial strategies. Larger spot sizes are more commonly employed for statistical fault attacks inferring secret data from device behavior across many attempts [[SFG⁺16](#)], while smaller ones are more often used for precise manipulation of data, e.g. flipping a single bit with high repeatability, to minimize the number of attempts needed to extract secrets [[DBC⁺18](#)]. However, when attacking implementations that are guaranteed to tolerate a certain threshold number of injected faults without revealing sensitive information, it can of course be beneficial to use a larger spot size to overcome this threshold [[BBM⁺22](#)]. Hence, the selection of the region has to be adapted to the relevant attacks on a certain target. The relationship between the laser spot size as well as its energy and the number of bits simultaneously faulted has been experimentally studied in works like [[DBC⁺18](#)]. Such experimental analyses are vital for understanding and defining effective fault regions and providing security in the region uniform random fault model in the most cost-effective manner. Electromagnetic fault injection attacks are commonly considered to be spatially localized as well, yet somewhat less so than laser fault attacks [[DLM19](#)]. Respective regions are likely to be similar in practice.

An orthogonal example of useful region groupings can be related to the clocking behavior of synchronous digital circuits. Combinational paths in a circuit-level implementation may be bundled by their total delay. When introducing an unexpectedly short cycle into the global clock signal (i.e., injecting a clock glitch) [[BCN⁺06](#)], the paths will be affected depending on the timing of the glitch relative to their delay. Clever attacks like fault sensitivity analysis may extract secret data by analyzing the data dependency of such behavior [[LSG⁺10](#)]. Finally, power glitching is another family of attacks, whose related

regions naturally also depend on path delay, as well as power domains. In summary, understanding how certain regions are affected by specific fault injection mechanisms is not a new question in fault analysis research. Characterization of device behavior under fault attacks has always been a common goal to design cost-effective countermeasures.

Next, let us discuss concrete instantiation of canary circuits. The *secret independence* of the canary circuits is a crucial requirement to avoid (statistical) ineffective fault attacks, therefore many fault detection schemes [SM12], such as those found in ineffective countermeasures [BSS20] are not suitable. A countermeasure similar to canary circuits is the use of analog fault injection sensors [HBB16, BBH17, EMV⁺24]: they are independent of the secrets and detect fault injection conditions. Therefore, they can be considered as a practical canary circuit instantiation and they can be integrated in a formal reduction similar to Theorem 3 if they are properly characterized. The *coverage* of each sensor should be identified, along with the *sensitivity*. Here the coverage means the kind of faults that are detected, formalized as regions of the circuit that virtually “contain” the sensor. The sensitivity is the probability of detecting the fault injection situation, as a function of the probability of injecting a fault in the circuit (i.e., of the parameter κ in the κ -random fault model). An ideal sensor would cover many regions of the circuit, while having a very high probability of detection as soon as κ is higher than a very low value. Our analysis framework however allows benefitting from less-than-ideal sensors, e.g. those that have blind spots [ANN24] by limiting the set of regions they cover (and therefore leaving those regions to other canaries).

Canaries could also be practically instantiated as digital circuits, closely matching our theoretical model. However, while the trivial canary compiler of Definition 5 achieves theoretical security, it uses only tie-low and tie-high gates in the canary circuit. A practical canary circuit will need more diversity to cover all the fault regions in the circuit. For example, values on some canary wires should change across clock cycles, and canary paths should have various delay to correspond to the clock glitching fault regions. Canary circuits should also represent the diversity of gates, driving strength, supply rails, paths, wires... contained in the actual circuit to protect and be evenly distributed within the physical layout (for laser/electromagnetic faults) regarding both the placement and the routing.

Finally, let us consider the *abort mechanism*. The instantiation in Section 4 demonstrates its theoretical feasibility, but it is practically very costly: its cost is $O(\ell n_o)$, where ℓ is the total number of canary wires and n_o is the number of circuit outputs. Since practical circuits run for many cycles, the value of ℓ is actually the number of physical canary wires multiplied by the number of clock cycles needed to evaluate the circuit, which would lead to a huge output gating circuit. A practical instantiation of this idea therefore needs to, at the very least, have some kind of compression scheme where the fault detection status is stored in a relatively low number of registers (in a fault-resilient way). While such a scheme might be practical, we can also consider a further optimization which is out of the scope of the theoretical circuit model: using the chip reset signal [low] as the abort mechanism. The problem then reduces to the design of a fault-resilient way of triggering a reset from the canary wires. Such an instantiation would also be beneficial against combined fault and side-channel attacks [AVFM07, RLK11]: by resetting the chip as soon as a fault is detected instead of only gating the output values, the side-channel leakage on faulty values is expected to be greatly reduced.